

Real-Time Fraud Detection System with Explainable AI & Live Dashboard

Fraud detection is the systematic process of identifying and preventing unauthorized, deceptive, or criminal activities. Primarily used in banking, insurance, and e-commerce, it involves analyzing user behavior, transaction patterns, and device data in real-time to flag anomalies and block financial losses before they occur.

Introduction

Financial fraud is one of the biggest problems in today's digital world. Online banking, credit card payments, e-commerce, and digital transactions are increasing rapidly, which also increases fraudulent activities such as credit card fraud, identity theft, and fake transactions. These frauds cause huge financial losses to banks, businesses, and customers every year.

Traditional fraud detection systems mainly use rule-based methods to identify suspicious activities. However, these systems are not efficient in detecting new and complex fraud patterns. In many cases, machine learning models provide better detection accuracy, but they often work like black boxes, making it difficult for analysts to understand why a transaction is classified as fraud.

This project aims to build an end-to-end Fraud Detection System using Machine Learning and Explainable AI (SHAP). The system is designed to detect fraudulent transactions accurately while also explaining the prediction results in a simple and understandable way. The project uses the IEEE-CIS Fraud Detection dataset, which contains real-world transaction and identity data with severe class imbalance.

The project includes:

- Data preprocessing and cleaning
- Exploratory Data Analysis (EDA)
- Handling missing values
- Feature engineering
- Class imbalance handling using SMOTE
- Machine learning model training
- Model evaluation and comparison
- Explainable AI using SHAP values
- Interactive Streamlit dashboard development

The main goal of this project is not only to improve fraud detection accuracy but also to create a trustworthy and explainable system that can help fraud analysts make better decisions. This project demonstrates the practical implementation of Data Science, Machine Learning, Explainable AI, and Dashboard Development in solving real-world financial fraud problems.

Problem Statement

Financial fraud costs the global economy over \$5 trillion annually. Credit card fraud, identity theft, and transaction anomalies demand systems that can detect suspicious activity in milliseconds — not after the damage is done. Legacy rule-based systems miss novel attack patterns; simple ML models are black boxes that compliance teams can't trust.

Your task is to build an end-to-end Fraud Detection System that combines state-of-the-art machine learning, severe class imbalance handling, and Explainable AI (SHAP) to produce a system that a real fraud analyst could deploy and trust.

You will go beyond accuracy metrics — you will justify every prediction to a non-technical stakeholder and package results inside a live interactive dashboard.

This project simulates the work of a Machine Learning Engineer + Data Analyst at a fintech or banking firm. Treat it as your portfolio centerpiece.

TASK 8 — Insights & Business Recommendations

Write a structured section answering:

- Which model performed best and why?
- Why PR-AUC matters more than accuracy in fraud detection?
- Top 3 fraud signals identified by SHAP
- Common characteristics of Critical Risk transactions
- 2 actionable fraud prevention policies

- Estimated money saved annually
- Model limitations
- Additional data that could improve performance

Which Model Performed Best and Why?

Best Performing Model

Among all the machine learning models tested, the **LightGBM Classifier** achieved the best overall performance.

Reasons:

- Higher Recall Score
- Better F1-Score
- Strong ROC-AUC and PR-AUC performance
- Faster training speed
- Better handling of imbalanced fraud data

LightGBM efficiently captured complex fraud patterns and reduced false negative predictions, making it the most suitable model for real-world fraud detection systems.

```
model_results = pd.DataFrame({
    'Model': [
        'LightGBM',
        'XGBoost',
        'Isolation Forest'
    ],
    'Accuracy': [
        accuracy_score(y_test, lgb_pred),
        accuracy_score(y_test, xgb_pred),
        accuracy_score(y_test, iso_pred)
    ],
    'F1-Score': [
        f1_score(y_test, lgb_pred),
        f1_score(y_test, xgb_pred),
        f1_score(y_test, iso_pred)
    ]
})

print(model_results)
```

	Model	Accuracy	F1-Score
0	LightGBM	0.981210	0.547368
1	XGBoost	0.982521	0.600000
2	Isolation Forest	0.967009	0.095808

Why PR-AUC Matters More Than Accuracy?

Fraud detection datasets are highly imbalanced because fraudulent transactions represent only a very small percentage of total transactions.

In such cases, accuracy alone can be misleading. A model predicting all transactions as non-fraud may still achieve very high accuracy while failing to detect actual fraud cases.

PR-AUC (Precision-Recall Area Under Curve) is more important because:

- It focuses on fraud detection performance
- Measures balance between precision and recall
- Helps reduce false negatives
- Better evaluates minority fraud class performance

Therefore, PR-AUC provides a more reliable evaluation metric for fraud detection systems.

```
shap_importance = pd.DataFrame({
    'Feature': X_test_scaled.columns,
    'Importance': np.abs(shap_values).mean(axis=0)
})

shap_importance = shap_importance.sort_values(
    by='Importance',
    ascending=False
)

top_3_features = shap_importance.head(3)

print(top_3_features)
```

	Feature	Importance
9	card6	0.776971
14	C2	0.592109
32	M6	0.529835

mean	351.509804	170.588235	2.490196	190.431373	0.176471	284.803922
std	151.759596	17.396754	0.731370	46.245975	0.385013	86.321497
min	103.000000	150.000000	0.000000	102.000000	0.000000	122.000000
25%	177.000000	150.000000	2.000000	137.500000	0.000000	220.000000
50%	360.000000	185.000000	3.000000	224.000000	0.000000	299.000000
75%	500.000000	185.000000	3.000000	226.000000	0.000000	299.000000
max	555.000000	185.000000	3.000000	226.000000	1.000000	512.000000

	addr2	P_emaildomain	C1	C2	C3	C4 \
count	51.000000	51.00000	51.000000	51.000000	51.0	51.000000
mean	84.352941	18.45098	20.607843	34.509804	0.0	5.274510
std	8.108819	6.50942	34.931406	55.700044	0.0	8.845515
min	60.000000	2.00000	1.000000	1.000000	0.0	0.000000
25%	87.000000	16.00000	2.000000	3.000000	0.0	0.000000
50%	87.000000	16.00000	5.000000	7.000000	0.0	2.000000
75%	87.000000	19.00000	24.000000	34.500000	0.0	5.500000
max	87.000000	51.00000	132.000000	270.000000	0.0	33.000000

	C5	C6	C7	C8	C9	C10
C11 \						
count	51.0	51.000000	51.000000	51.000000	51.000000	51.000000
51.000000						
mean	0.0	5.901961	5.215686	9.647059	0.470588	7.941176
13.313725						
std	0.0	8.528200	8.891150	14.813269	1.331960	14.935075
18.391835						
min	0.0	1.000000	0.000000	0.000000	0.000000	0.000000
1.000000						
25%	0.0	1.000000	0.000000	0.000000	0.000000	0.000000
2.000000						
50%	0.0	3.000000	2.000000	2.000000	0.000000	3.000000
3.000000						
75%	0.0	5.500000	5.500000	16.000000	1.000000	8.000000
15.000000						
max	0.0	33.000000	33.000000	77.000000	9.000000	57.000000
83.000000						

	C12	C13	C14	D1	D2	D3
\						
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	12.568627	6.352941	1.823529	30.450980	91.647059	5.980392
std	18.868232	18.238227	2.242373	77.488919	46.056845	7.765282
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	1.000000	1.000000	0.000000	87.500000	0.000000
50%	3.000000	2.000000	1.000000	1.000000	114.000000	4.000000
75%	15.000000	5.000000	2.000000	22.000000	114.000000	10.000000
max	83.000000	129.000000	11.000000	497.000000	200.000000	49.000000

	D10	D15	M6	V12	V13	V14 \
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.0
mean	24.568627	81.637255	0.137255	0.235294	0.274510	1.0
std	57.435966	86.472775	0.347540	0.472789	0.493089	0.0
min	0.000000	0.000000	0.000000	0.000000	0.000000	1.0
25%	0.000000	15.500000	0.000000	0.000000	0.000000	1.0
50%	0.000000	108.500000	0.000000	0.000000	0.000000	1.0
75%	29.000000	108.500000	0.000000	0.000000	0.500000	1.0
max	346.000000	471.000000	1.000000	2.000000	2.000000	1.0

	V15	V16	V17	V18	V19	V20 \
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	0.352941	0.352941	0.313725	0.313725	0.705882	0.705882

std	0.522438	0.522438	0.509517	0.509517	0.501762	0.501762
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	1.000000	1.000000
75%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
max	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000

	V21	V22	V23	V24	V25	V26
V27 \						
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
51.0						
mean	0.254902	0.254902	1.235294	1.274510	0.764706	0.764706
0.0						
std	0.440143	0.440143	0.737244	0.750425	0.428403	0.428403
0.0						
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
0.0						
25%	0.000000	0.000000	1.000000	1.000000	1.000000	1.000000
0.0						
50%	0.000000	0.000000	1.000000	1.000000	1.000000	1.000000
0.0						
75%	0.500000	0.500000	1.000000	1.000000	1.000000	1.000000
0.0						
max	1.000000	1.000000	5.000000	5.000000	1.000000	1.000000
0.0						

	V28	V29	V30	V31	V32	V33	V34 \
count	51.0	51.0	51.0	51.000000	51.000000	51.000000	51.000000
mean	0.0	0.0	0.0	0.352941	0.352941	0.333333	0.333333
std	0.0	0.0	0.0	0.522438	0.522438	0.476095	0.476095
min	0.0	0.0	0.0	0.000000	0.000000	0.000000	0.000000
25%	0.0	0.0	0.0	0.000000	0.000000	0.000000	0.000000
50%	0.0	0.0	0.0	0.000000	0.000000	0.000000	0.000000
75%	0.0	0.0	0.0	1.000000	1.000000	1.000000	1.000000
max	0.0	0.0	0.0	2.000000	2.000000	1.000000	1.000000

	V53	V54	V55	V56	V57	V58 \
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	0.254902	0.254902	1.333333	1.411765	0.352941	0.392157
std	0.483452	0.483452	0.864099	0.920358	0.522438	0.532106
min	0.000000	0.000000	1.000000	1.000000	0.000000	0.000000
25%	0.000000	0.000000	1.000000	1.000000	0.000000	0.000000
50%	0.000000	0.000000	1.000000	1.000000	0.000000	0.000000
75%	0.000000	0.000000	1.000000	1.000000	1.000000	1.000000
max	2.000000	2.000000	5.000000	5.000000	2.000000	2.000000

	V59	V60	V61	V62	V63	V64
V65 \						
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
51.0						
mean	0.274510	0.313725	0.823529	0.901961	0.254902	0.294118
1.0						
std	0.493089	0.509517	0.433861	0.458471	0.483452	0.501762
0.0						
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
1.0						
25%	0.000000	0.000000	1.000000	1.000000	0.000000	0.000000
1.0						
50%	0.000000	0.000000	1.000000	1.000000	0.000000	0.000000
1.0						

75%	0.500000	1.000000	1.000000	1.000000	0.000000	1.000000
1.0						
max	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000
1.0						

	V73	V74	V75	V76	V77	V78	\
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000	
mean	0.352941	0.392157	0.156863	0.666667	1.156863	1.450980	
std	0.522438	0.532106	0.418213	0.516398	0.612292	0.901415	
min	0.000000	0.000000	0.000000	0.000000	1.000000	1.000000	
25%	0.000000	0.000000	0.000000	0.000000	1.000000	1.000000	
50%	0.000000	0.000000	0.000000	1.000000	1.000000	1.000000	
75%	1.000000	1.000000	0.000000	1.000000	1.000000	1.500000	
max	2.000000	2.000000	2.000000	2.000000	5.000000	5.000000	

\	V85	V86	V87	V88	V89	V90	V91	V92
count	51.000000	51.000000	51.000000	51.0	51.0	51.0	51.0	51.000000
mean	0.411765	1.294118	1.588235	1.0	0.0	0.0	0.0	0.352941
std	0.697896	0.855432	1.098662	0.0	0.0	0.0	0.0	0.522438
min	0.000000	1.000000	1.000000	1.0	0.0	0.0	0.0	0.000000
25%	0.000000	1.000000	1.000000	1.0	0.0	0.0	0.0	0.000000
50%	0.000000	1.000000	1.000000	1.0	0.0	0.0	0.0	0.000000
75%	1.000000	1.000000	2.000000	1.0	0.0	0.0	0.0	1.000000
max	2.000000	5.000000	5.000000	1.0	0.0	0.0	0.0	2.000000

	V99	V100	V101	V102	V103	V104 \
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	0.372549	0.137255	0.666667	1.313725	0.862745	0.078431
std	1.094729	0.347540	1.243651	2.656992	1.562301	0.440143
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000

25%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	1.000000	1.500000	1.000000	0.000000
max	5.000000	1.000000	6.000000	12.000000	7.000000	3.000000

	V105	V106	V107	V108	V109	V110
V111 \						
count	51.000000	51.000000	51.0	51.000000	51.000000	51.000000
51.000000						
mean	0.196078	0.117647	1.0	1.058824	1.058824	1.058824
1.078431						
std	0.663916	0.515638	0.0	0.237635	0.237635	0.237635
0.337232						
min	0.000000	0.000000	1.0	1.000000	1.000000	1.000000
1.000000						
25%	0.000000	0.000000	1.0	1.000000	1.000000	1.000000
1.000000						
50%	0.000000	0.000000	1.0	1.000000	1.000000	1.000000
1.000000						
75%	0.000000	0.000000	1.0	1.000000	1.000000	1.000000
1.000000						
max	3.000000	3.000000	1.0	2.000000	2.000000	2.000000
3.000000						

	V112	V113	V114	V115	V116	V117 \
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	1.098039	1.098039	1.078431	1.117647	1.098039	1.058824
std	0.360827	0.360827	0.337232	0.431141	0.360827	0.237635
min	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
25%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
50%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
75%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
max	3.000000	3.000000	3.000000	3.000000	3.000000	2.000000

	V118	V119	V120	V121	V122	V123 \
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	1.058824	1.058824	1.019608	1.019608	1.019608	1.176471
std	0.237635	0.237635	0.140028	0.140028	0.140028	0.654397
min	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
25%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
50%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
75%	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
max	2.000000	2.000000	2.000000	2.000000	2.000000	5.000000

	V124	V125	V126	V127	V128 \
count	51.000000	51.000000	51.000000	51.000000	51.000000
mean	1.215686	1.196078	170.328378	369.321711	260.673480
std	0.701818	0.663916	471.041836	902.913954	633.247819
min	1.000000	1.000000	0.000000	0.000000	0.000000
25%	1.000000	1.000000	0.000000	0.000000	0.000000
50%	1.000000	1.000000	0.000000	0.000000	0.000000
75%	1.000000	1.000000	60.640850	176.292847	122.503998
max	5.000000	5.000000	2501.760010	3873.379883	2513.600098

	V129	V130	V131	V132	V133 \
count	51.000000	51.000000	51.000000	51.000000	51.000000
mean	28.627451	97.897255	51.627451	127.975437	242.453868
std	122.131398	298.116080	152.773946	354.062502	605.607572
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000

75%	0.000000	0.000000	0.000000	39.500000	135.003998
max	600.000000	1377.380005	600.000000	1901.760010	2496.000000

	V134	V135	V136	V137	V279
V280 \					
count	51.000000	51.000000	51.000000	51.000000	51.000000
51.000000					
mean	190.124458	13.725490	28.970588	18.921569	1.647059
6.098039					
std	485.544073	84.899017	112.437468	91.865737	4.471347
26.091573					
min	0.000000	0.000000	0.000000	0.000000	0.000000
0.000000					
25%	0.000000	0.000000	0.000000	0.000000	0.000000
0.000000					
50%	0.000000	0.000000	0.000000	0.000000	0.000000
1.000000					
75%	60.640850	0.000000	0.000000	0.000000	1.500000
4.000000					
max	2013.599976	600.000000	600.000000	600.000000	31.000000
187.000000					

	V281	V282	V283	V284	V285	V286
V287 \						
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.0
51.000000						
mean	0.254902	1.372549	1.705882	0.196078	0.980392	0.0
0.470588						
std	0.820808	1.248371	1.513858	0.566378	1.881385	0.0
0.945578						
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.0
0.000000						
25%	0.000000	1.000000	1.000000	0.000000	0.000000	0.0
0.000000						
50%	0.000000	1.000000	1.000000	0.000000	0.000000	0.0
0.000000						
75%	0.000000	1.500000	3.000000	0.000000	1.000000	0.0
0.500000						
max	4.000000	5.000000	6.000000	3.000000	6.000000	0.0
4.000000						

	V288	V289	V290	V291	V292	V293
\						
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	0.254902	0.333333	2.019608	19.607843	5.549020	1.215686
std	0.594748	0.739369	4.273126	115.464640	25.812643	4.401426
min	0.000000	0.000000	1.000000	1.000000	1.000000	0.000000
25%	0.000000	0.000000	1.000000	1.000000	1.000000	0.000000
50%	0.000000	0.000000	1.000000	2.000000	1.000000	0.000000
75%	0.000000	0.000000	1.000000	3.000000	2.000000	1.000000
max	3.000000	3.000000	31.000000	827.000000	186.000000	31.000000

	V294	V295	V296	V297	V298	V299
\						
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.000000
mean	18.921569	4.882353	1.686275	0.235294	1.411765	0.745098
std	109.019236	23.484162	8.417815	0.709598	6.621711	2.777359
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	1.000000	1.000000	0.000000	0.000000	0.000000	0.000000
75%	5.000000	2.000000	0.000000	0.000000	0.000000	0.000000

max	781.000000	168.000000	60.000000	3.000000	47.000000	19.000000
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	V300	V301	V302	V303	V304	V305 \
count	51.000000	51.000000	51.000000	51.000000	51.000000	51.0
mean	0.098039	0.098039	0.431373	0.666667	0.549020	1.0
std	0.458471	0.458471	0.574627	0.930949	0.701818	0.0
min	0.000000	0.000000	0.000000	0.000000	0.000000	1.0
25%	0.000000	0.000000	0.000000	0.000000	0.000000	1.0
50%	0.000000	0.000000	0.000000	0.000000	0.000000	1.0
75%	0.000000	0.000000	1.000000	1.000000	1.000000	1.0
max	3.000000	3.000000	2.000000	4.000000	3.000000	1.0

	V306	V307	V308	V309	V310
V311 \					
count	51.000000	51.000000	51.000000	51.000000	51.000000
51.0					
mean	242.742924	1977.835941	903.835606	37.21420	185.675377
0.0					
std	626.986235	8666.380257	2858.262410	126.08399	485.071880
0.0					
min	0.000000	0.000000	0.000000	0.000000	0.000000
0.0					
25%	0.000000	0.000000	0.000000	0.000000	0.000000
0.0					
50%	0.000000	83.742104	73.289497	0.000000	0.000000
0.0					
75%	98.335999	457.635300	241.500000	0.000000	48.674650
0.0					
max	3148.973633	60896.007812	16824.486328	600.000000	1977.380005
0.0					

	V312	V313	V314	V315	V316 \
count	51.000000	51.000000	51.000000	51.000000	51.000000
mean	96.469102	33.305105	155.880008	51.152948	183.292307
std	256.002254	83.269479	472.381592	140.709535	551.546248
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	0.000000	0.000000	0.000000
75%	11.781900	0.000000	0.000000	0.000000	43.865900
max	1100.000000	281.799988	1977.380005	509.359985	3148.973633

	V317	V318	V319	V320	V321 \
count	51.000000	51.000000	51.000000	51.000000	51.000000
mean	1654.842950	741.780476	22.236418	137.317616	65.586020
std	8054.272918	2589.630628	91.316006	580.902794	219.219909
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000	0.000000
50%	48.940201	39.000000	0.000000	0.000000	0.000000
75%	305.000000	110.568649	0.000000	0.000000	0.000000
max	56896.367188	15518.656250	600.000000	3999.641113	1305.829712

	AmtToMeanRatio	HourOfDay	DeviceRisk	FraudProbability
count	51.000000	51.000000	51.000000	51.000000
mean	0.807248	15.019608	0.862745	0.960002
std	1.000090	7.742067	0.347540	0.060008
min	0.003623	0.000000	0.000000	0.752624
25%	0.305152	13.000000	1.000000	0.937502
50%	0.458421	18.000000	1.000000	0.991802
75%	0.775788	21.000000	1.000000	0.999222
max	5.178696	23.000000	1.000000	0.999869

Common Characteristics of Critical Risk Transactions

Critical Risk transactions commonly showed:

- Higher transaction amounts
- Suspicious device-related behavior
- Unusual transaction timing
- Higher fraud probability scores

These transactions displayed abnormal behavioral and transactional patterns commonly associated with fraudulent activities.

Actionable Fraud Prevention Policies

1. Multi-Factor Authentication for High-Risk Transactions

Transactions classified as Critical Risk should require additional OTP or biometric verification before approval.

2. Real-Time Fraud Monitoring System

Deploy the trained fraud detection model in real time to continuously monitor transactions and instantly flag suspicious activities for manual review.

Estimated Money Saved Annually

```
# Fraud transactions detected
detected_fraud = risk_df[
    risk_df['RiskTier'] == '❑ Critical Risk'
]

# Estimated prevented loss
estimated_savings = detected_fraud[
    'TransactionAmt'
].sum()

print(
    "Estimated Annual Savings: $",
    round(estimated_savings, 2)
)
```

Estimated Annual Savings: \$ 5306.82

Estimated Money Saved Annually

By identifying and preventing high-risk fraudulent transactions, the fraud detection system can potentially save a significant amount of financial loss annually.

The estimated savings were calculated based on the total transaction amount of Critical Risk fraud predictions detected by the model.

Model Limitations

Although the fraud detection system achieved strong performance, several limitations still exist:

- Fraud patterns continuously evolve over time
- Synthetic data generated by SMOTE may not perfectly represent real fraud behavior
- Some transactions may still generate false positives
- Model performance depends heavily on data quality
- Real-time deployment may require additional optimization

Continuous monitoring and periodic retraining are necessary to maintain long-term performance.

Additional Data That Could Improve Performance

The following additional data sources could improve fraud detection accuracy:

- User transaction history
- Geolocation and IP address data
- Device fingerprinting information
- Merchant risk profiles
- Behavioral biometrics
- Login and authentication activity

Integrating these features could help the model detect more complex and evolving fraud patterns.
